



DESCRIPTIVE STATISTICS

Foundations for Data Science and Machine Learning

COURSE ROADMAP

Six sections — each ends with  exercises and  solutions

01

Introduction
& Variables

02

Frequency
Tables

03

Central
Tendency

04

Spread &
Variability

05

Distributions

06

Summary &
Applications

 = Practice Exercises

 = Step-by-Step Solutions

Each section: Concepts →  Exercises →  Full Solutions

SECTION 01

INTRODUCTION & VARIABLES

What is statistics, why it matters, and the data types we work with

WHAT IS STATISTICS?

The science of collecting, analyzing, interpreting, and presenting data

[STATISTICS]

The science concerned with developing and studying methods for collecting, analyzing, interpreting and presenting empirical data. It is a highly interdisciplinary field — applicable in virtually all scientific domains.

UNCERTAINTY

Statistics deals with uncertainty and variation — two fundamental ideas that underpin every statistical method.

VARIATION

Data values differ across observations. Statistics quantifies and models this variation to draw conclusions.

INTERDISCIPLINARY

Research questions in science motivate new statistical methods, which are then applied back to those fields.

TOOLS USED

Mathematical and computational tools: probability, calculus, linear algebra, and modern computing.

STATISTICS IN DATA SCIENCE

Core concepts and terminology every data scientist must know

In data science, statistics is at the core of machine learning algorithms — capturing and translating data patterns into actionable evidence. Data scientists use statistics to gather, review, analyse, and draw conclusions from data.

Population

The full source of data to be studied — all possible observations.

Sample

A representative subset selected from the population.

Variable

Any data item that can be measured or counted.

Descriptive Stats

Characteristics that summarize a population or sample.

Inferential Stats

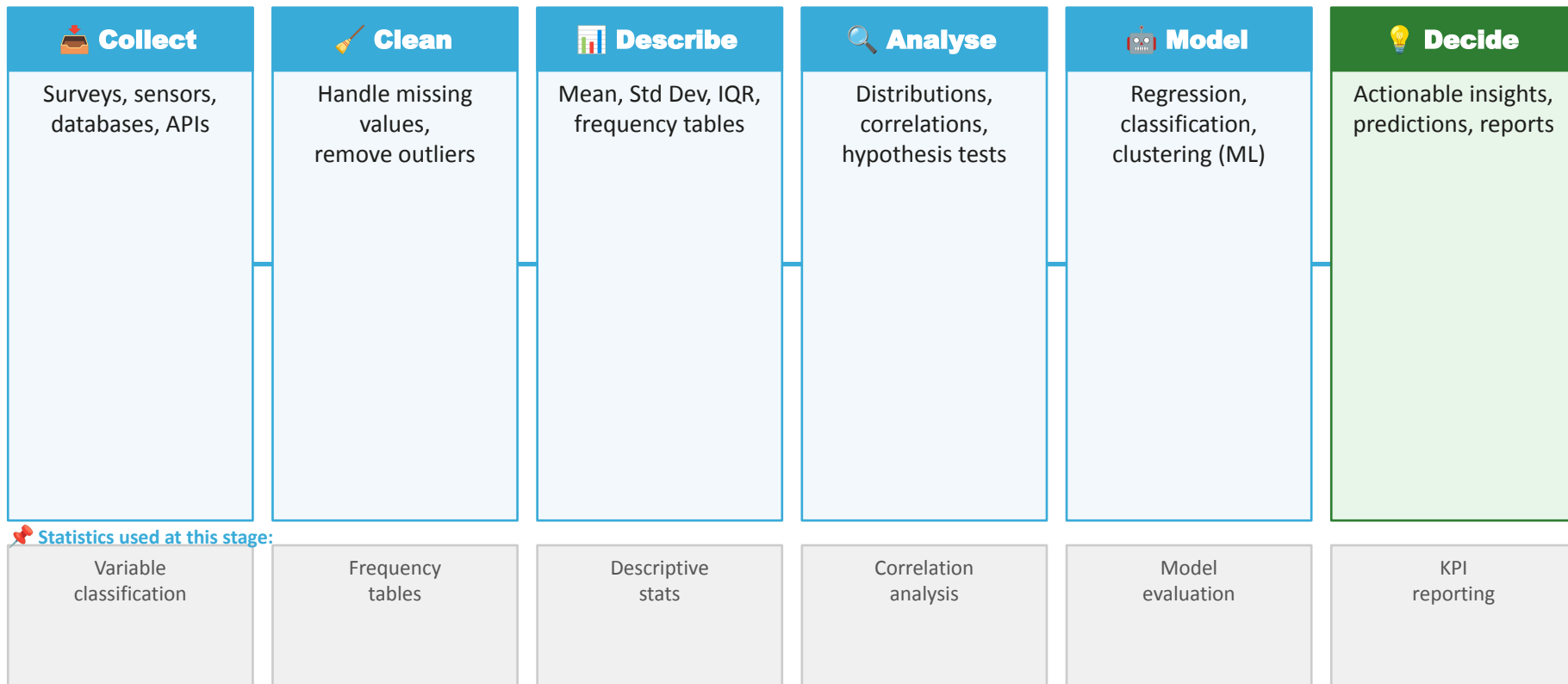
Predictions and conclusions drawn about a population from a sample.

Quantitative Analysis

Collecting and interpreting data with patterns and visualizations.

STATISTICS IN DATA SCIENCE

How statistics powers every stage of the data pipeline



VARIABLES

Any characteristic, number, or quantity that can be measured or counted

[VARIABLE]

A variable is any characteristic, number, or quantity that can be measured or counted. It may vary between data units in a population or change in value over time.

Examples: Age, sex, income, eye colour, temperature, class grade.

TYPES OF VARIABLES:

NUMERIC (QUANTITATIVE)

◆ Continuous

Can take ANY value in a range.
Examples: height, time, temperature, age.
Measured — not counted.

◆ Discrete

Takes only WHOLE number values.
Examples: # of cars, # of children.
Counted — not measured.

CATEGORICAL (QUALITATIVE)

◆ Ordinal

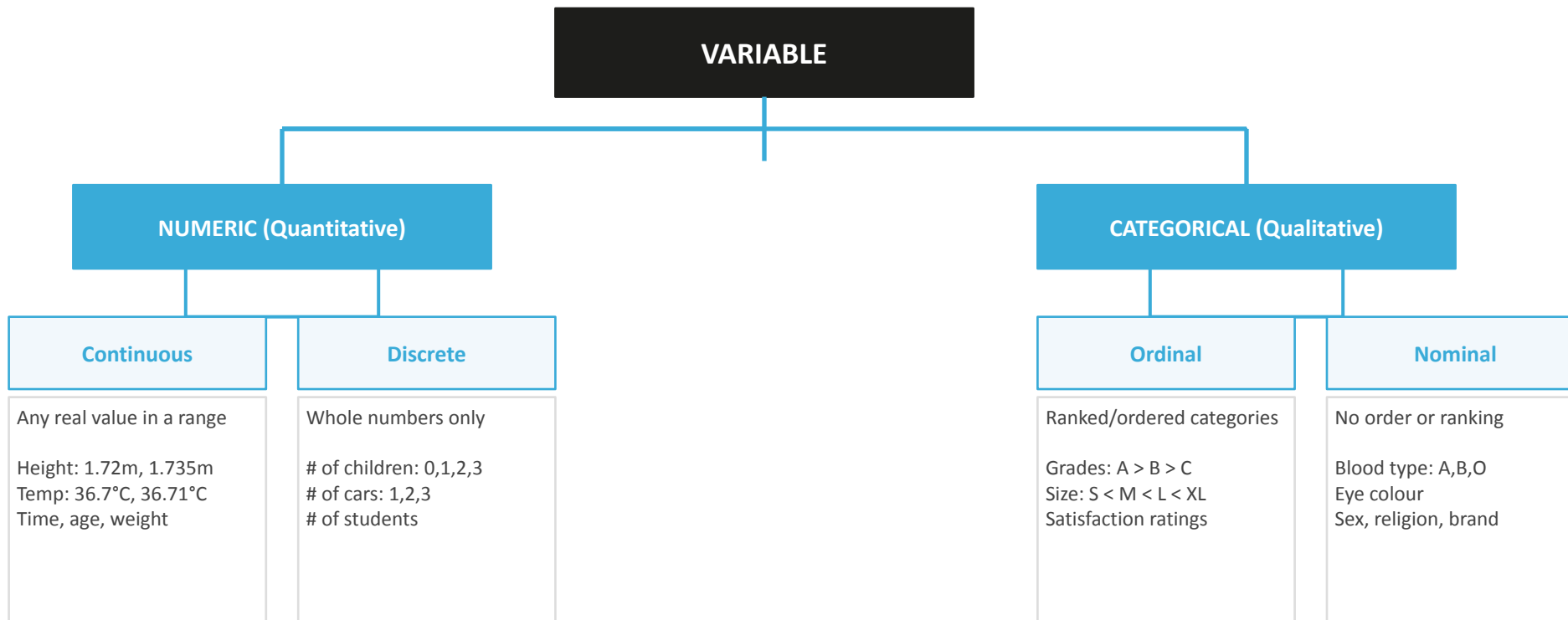
Can be RANKED or ordered.
Examples: grades (A,B,C), clothing size (S,M,L), satisfaction (agree/disagree).

◆ Nominal

NO logical order or ranking.
Examples: eye colour, religion, sex, brand, country.

TYPES OF VARIABLES — VISUAL OVERVIEW

The complete classification tree



PRACTICE EXERCISES

SECTION 01 — INTRODUCTION & VARIABLES

 TRY IT

Exercise 1 — Classify the Variable

For each item below, state whether it is Numeric-Continuous, Numeric-Discrete, Categorical-Ordinal, or Categorical-Nominal:

- a) Body temperature (in °C) b) Number of siblings c) Movie rating (1–5 stars) d) Blood type (A, B, AB, O)

Exercise 2 — Population vs Sample

A researcher wants to study the average screen time of university students in Egypt.

- a) What is the population? b) Describe a reasonable sample. c) Why might the sample differ from the population?

Exercise 3 — Key Definitions

Explain the difference between Descriptive Statistics and Inferential Statistics using a concrete example involving student exam scores.

✓ Solution 1 — Classify the Variable

Step 1: a) Body temperature → Numeric-Continuous (takes any real value, e.g. 36.7°C, 37.2°C)

Step 2: b) Number of siblings → Numeric-Discrete (whole numbers only: 0, 1, 2, ...)

Step 3: c) Movie rating (1–5 stars) → Categorical-Ordinal (ordered ranks, but gaps between ratings aren't necessarily equal)

Step 4: d) Blood type → Categorical-Nominal (categories with no logical order)

✓ Solution 2 — Population vs Sample

Step 1: a) Population: ALL university students in Egypt

Step 2: b) Sample: e.g. 500 randomly selected students from 5 universities across different regions

Step 3: c) The sample may differ due to sampling bias, non-response, or natural variation — that's why we use statistical inference to generalise carefully

✓ Solution 3 — Descriptive vs Inferential

Step 1: Descriptive Statistics summarises the data you have — e.g. 'The average exam score of our 40-student class is 72, with a standard deviation of 8.'

Step 2: Inferential Statistics uses that sample to make predictions about a larger group — e.g. 'Based on our class, we estimate the average score for all students in this course is between 70 and 74 (95% confidence).'

SECTION 02

FREQUENCY TABLES

Organising, counting, and understanding how data values distribute

FREQUENCY TABLES

How often does each value appear in your dataset?

[FREQUENCY TABLE]

A table that lists each distinct value in a dataset alongside the count (frequency) of how often it appears. Also called a frequency distribution — it reveals which values are common and which are rare.

Frequency

The raw count of how many times a value appears in the dataset.

Relative Frequency

Frequency $\div$ Total observations. Expressed as a proportion or %.

Cumulative Frequency

Running total of frequencies from the lowest to a given value.

Class / Bin

For continuous data, values are grouped into intervals (bins).

FREQUENCY TABLE — WORKED EXAMPLE

Building a table step by step

Dataset: Exam scores of 10 students: 55, 70, 70, 85, 90, 70, 55, 85, 90, 90

◆ Step 1 — List unique values & count

Unique values: 55, 70, 85, 90

Count each: 55 → 2 times, 70 → 3 times, 85 → 2 times, 90 → 3 times

Total observations (n) = 10

Score	Frequency	Rel. Freq.	Cum. Freq.
55	2	20%	2
70	3	30%	5
85	2	20%	7
90	3	30%	10
Total	10	100%	

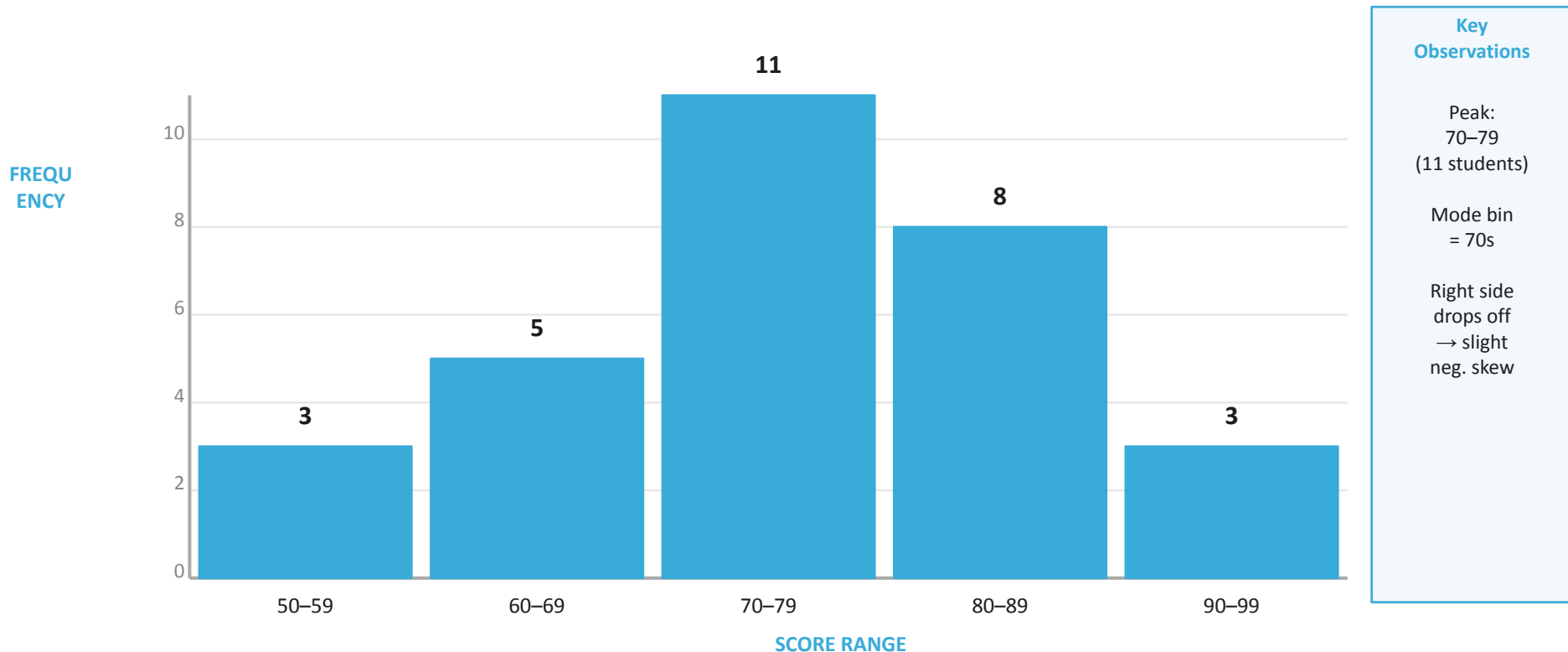
HOW TO READ THIS TABLE

- Frequency: The score 70 appeared 3 times in the dataset.
- Relative Frequency: 70 represents 30% of all scores (3/10).
- Cumulative Frequency: By the time we reach 85, we've accounted for 7 out of 10 scores (70%).
- Use: Instantly see that 90 is the most common score (tied with 70), and 55 is the least common.

FREQUENCY HISTOGRAM

Visualising how data values distribute across a range

Dataset: Exam scores of 30 students grouped into bins of 10.



PRACTICE EXERCISES

SECTION 02 — FREQUENCY TABLES



Exercise 1 — Build a Frequency Table

Dataset — Daily steps walked by 12 people: 4000, 7000, 5000, 7000, 9000, 5000, 7000, 4000, 9000, 5000, 7000, 9000

Build a complete frequency table with: Value, Frequency, Relative Frequency (%), Cumulative Frequency.

Exercise 2 — Interpret a Frequency Table

A frequency table shows: Grade A $\rightarrow$ 5 students, Grade B $\rightarrow$ 12 students, Grade C $\rightarrow$ 8 students, Grade D $\rightarrow$ 3 students, Grade F $\rightarrow$ 2 students (Total = 30)

a) What % of students passed (grade A, B, or C)? b) What is the cumulative frequency up to and including grade B?

Exercise 3 — Relative Frequency

In a survey of 200 people: 80 prefer coffee, 60 prefer tea, 40 prefer juice, 20 prefer water.

a) Compute the relative frequency for each drink. b) Which drink has the highest relative frequency?

SOLUTIONS (1 of 2)

SECTION 02 — FREQUENCY TABLES

✓ ANSWERS

✓ Solution 1 — Build a Frequency Table

Step 1: Step 1 — Unique values: 4000, 5000, 7000, 9000

Step 2: Step 2 — Count: 4000→2, 5000→3, 7000→4, 9000→3 (Total $n=12$)

Step 3: Step 3 — Rel. Freq.: $4000=2/12=17\%$, $5000=3/12=25\%$, $7000=4/12=33\%$, $9000=3/12=25\%$

Step 4: Step 4 — Cum. Freq.: 4000→2, 5000→5, 7000→9, 9000→12

Step 5: Interpretation: 7000 steps is the most common value (mode) at 33% of observations.

SOLUTIONS (2 of 2)

SECTION 02 — FREQUENCY TABLES

✓ ANSWERS

✓ Solution 2 — Interpret a Frequency Table

Step 1: a) Passing grades (A+B+C) = $5+12+8 = 25$ students $\rightarrow 25/30 = 83.3\%$ passed ✓

Step 2: b) Cumulative frequency up to grade B = students with A + students with B = $5+12 = 17$ students

✓ Solution 3 — Relative Frequency

Step 1: Step 1 — Total = 200

Step 2: Coffee: $80/200 = 40\%$ Tea: $60/200 = 30\%$ Juice: $40/200 = 20\%$ Water: $20/200 = 10\%$

Step 3: Check: $40+30+20+10 = 100\%$ ✓

Step 4: Coffee has the highest relative frequency at 40%.

SECTION 03

CENTRAL TENDENCY

Mean, Median, and Mode — three ways to find the 'typical' value

MEAN, MEDIAN & MODE

Three measures that summarize a dataset with a single number

Each measure tries to represent the 'typical' data point — but they respond differently to outliers and data shape.

MEAN (Average)

$$\bar{x} = (\sum x_i) / n$$

Sum all values and divide by count.
Affected by outliers — one extreme value pulls the mean.
Best for symmetric, outlier-free data.

◆ Data: 4, 1, 7
Mean = $(4+1+7)/3 = 12/3 = 4$

MEDIAN (Middle)

Middle value when sorted

Sort all values; pick the middle one.
If even count: average the two middle values.
More reliable when outliers are present.

◆ Data: 1, 4, 7
Median = 4 (middle value)

MODE (Most Frequent)

Value with highest frequency

The value that appears most often.
A dataset can have 0, 1, or multiple modes.
Used for categorical data.

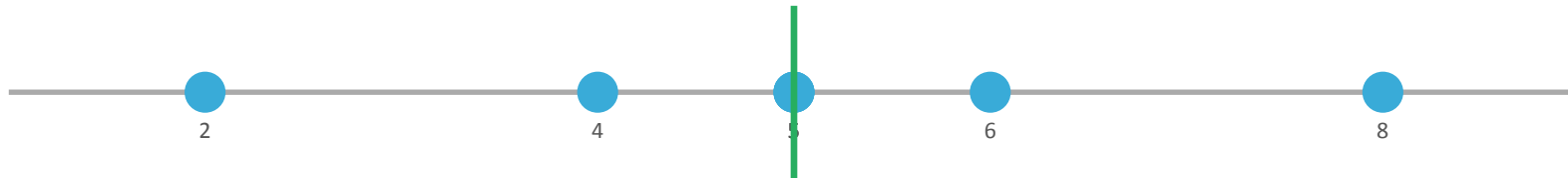
◆ Data: {4, 2, 4, 3, 2, 2}
Mode = 2 (appears 3 times)

MEAN, MEDIAN & MODE — VISUAL COMPARISON

How outliers shift the three measures apart

SYMMETRIC DATASET: {2, 4, 5, 5, 5, 6, 8} Mean = Med

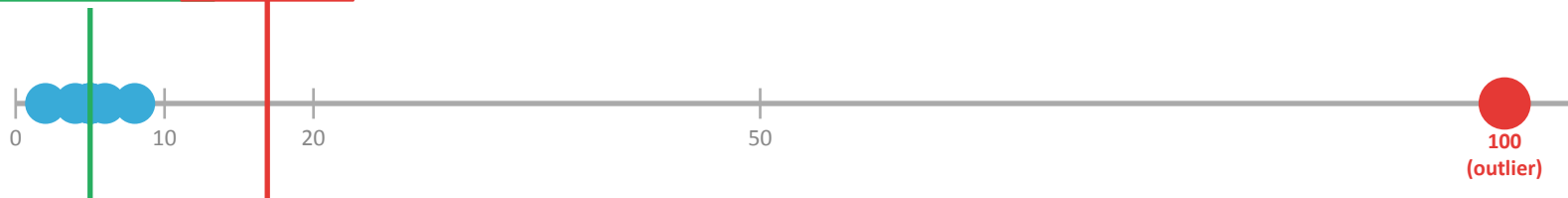
Mean = Median = Mode = 5



DATASET WITH OUTLIER: {2, 4, 5, 5, 5, 6, 8, 100} Mean=16.9 | Median=5 | Mode=5

Mode=Median=5

Mean=16.9



KEY INSIGHT: Adding one outlier (100) barely moves the Median (stays at 5) but triples the Mean (from 5 → 16.9). This is why the Median is the preferred measure for skewed data like salaries and house prices.

CHOOSING THE RIGHT MEASURE

When to use Mean, Median, or Mode

Situation	Best Measure	Reason
Symmetric data, no outliers	Mean	Uses all data values — most precise
Data with extreme outliers (e.g. salaries)	Median	Not affected by very high or very low values
Categorical data (e.g. eye colour)	Mode	Mean & Median don't apply to categories
Skewed distribution	Median	Mean is pulled toward the tail
Most common value needed	Mode	Directly gives highest-frequency value
Missing data imputation	Median	Resistant to distortion from other values

 **OUTLIER EFFECT:** Data = {2, 3, 4, 5, 100}
Mean = 22.8 | Median = 4 | Mode = none
The single outlier (100) drags the mean far from the typical value. Median stays robust.

PRACTICE EXERCISES

SECTION 03 — CENTRAL TENDENCY

 TRY IT

Exercise 1 — Calculate Mean, Median, and Mode

Dataset: exam scores of 9 students: 55, 70, 85, 70, 90, 55, 70, 80, 95

- a) Calculate the Mean b) Find the Median (show sorting step) c) Identify the Mode

Exercise 2 — Effect of Outliers

Original dataset: {10, 12, 11, 13, 12, 14, 10}

- a) Compute Mean and Median of this dataset.
b) Now add one outlier — 200 — to the dataset. Recompute Mean and Median.
c) Which measure was more affected? What does this tell you?

Exercise 3 — Choose the Right Measure

For each scenario, state which measure of central tendency is most appropriate and explain why:

- a) Reporting the 'typical' house price in Cairo (some mansions worth 50M EGP exist)
b) The most popular car colour sold at a dealership
c) The average height of players on a basketball team

✓ Solution 1 — Mean, Median & Mode

Step 1: Sorted data: 55, 55, 70, 70, 70, 80, 85, 90, 95 ($n = 9$)

Step 2: Mean = $(55+55+70+70+70+80+85+90+95) / 9 = 670/9 \approx 74.4$

Step 3: Median = middle value of sorted list = $9\text{th}/2 = 5\text{th value} = 70$

Step 4: Mode = 70 (appears 3 times — more than any other value)

✓ Solution 2 — Effect of Outliers

Step 1: Original {10,12,11,13,12,14,10}: Mean = $(10+12+11+13+12+14+10)/7 = 82/7 \approx 11.7$. Sorted: 10,10,11,12,12,13,14 → Median = 12

Step 2: With outlier 200 added ($n=8$): Sum = 282 → Mean = $282/8 = 35.25$. Sorted: 10,10,11,12,12,13,14,200 → Median = $(12+12)/2 = 12$

Step 3: Mean jumped from 11.7 to 35.25 (200% increase!) while Median stayed at 12 → Mean is far more sensitive to outliers

✓ Solution 3 — Choose the Right Measure

Step 1: a) House prices in Cairo → MEDIAN. A few mansions worth 50M EGP will massively inflate the mean, making it unrepresentative of typical buyers.

Step 2: b) Most popular car colour → MODE. This is categorical data — mean and median are meaningless for colours.

Step 3: c) Basketball team heights → MEAN. Heights are numeric, relatively symmetric, and no extreme outliers expected — mean uses all information.

SECTION 04

SPREAD & VARIABILITY

Variance, Standard Deviation, Range, and IQR — how spread out is the data?

VARIANCE (σ^2)

How far are the data points scattered from the mean?

[VARIANCE σ^2]

A measure of how spread out data points are from their mean. Always non-negative (squares eliminate sign). Has squared units — cannot be compared directly to the mean.

$$\text{Population: } \sigma^2 = \sum (x_i - \mu)^2 / N$$

$$\text{Sample: } s^2 = \sum (x_i - \bar{x})^2 / (n-1)$$

Always ≥ 0

Squaring every deviation means variance can never be negative.

Squared Units

If data is in kg, variance is in kg^2 . Cannot compare directly to mean.

Low = Clustered

A variance near 0 means data points are very close to the mean.

High = Spread

A large variance means data points are widely scattered.

STANDARD DEVIATION (σ)

The square root of variance — same units as the original data

[STD DEVIATION σ]

The square root of the variance. Measures the average distance of each data point from the mean. Unlike variance, it has the SAME units as the data — directly comparable to the mean.

$$\text{Population: } \sigma = \sqrt{[\sum (x_i - \mu)^2 / N]}$$

$$\text{Sample: } s = \sqrt{[\sum (x_i - \bar{x})^2 / (n-1)]}$$

◆ Step-by-Step Example — Dataset: {2, 4, 4, 4, 5, 5, 7, 9}

Step 1: Mean = $(2+4+4+4+5+5+7+9)/8 = 40/8 = 5$

Step 2: Squared deviations: $(2-5)^2=9$, $(4-5)^2=1$, $(4-5)^2=1$, $(4-5)^2=1$, $(5-5)^2=0$, $(5-5)^2=0$, $(7-5)^2=4$, $(9-5)^2=16$

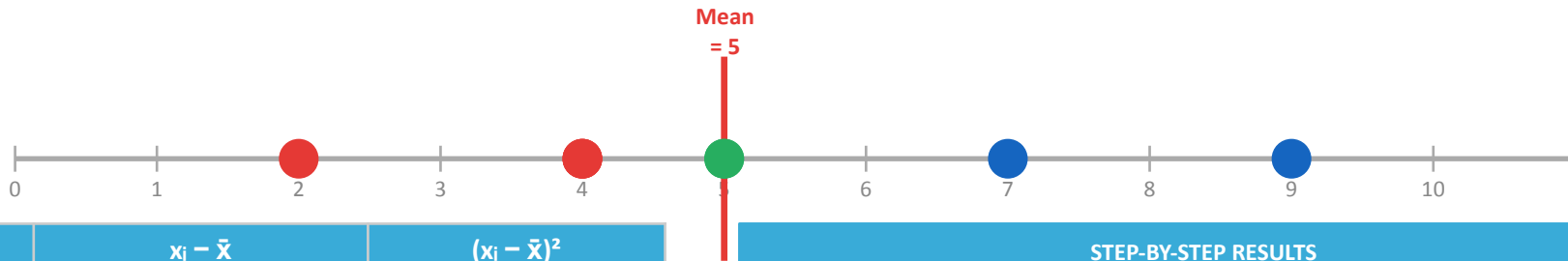
Step 3: Variance $\sigma^2 = (9+1+1+1+0+0+4+16)/8 = 32/8 = 4$

Step 4: Std Dev $\sigma = \sqrt{4} = 2 \rightarrow$ On average each value is 2 units from the mean.

VARIANCE & STD DEVIATION — VISUAL

Why we square the deviations, and what the result means

Dataset: {2, 4, 4, 4, 5, 5, 7, 9} Mean = 5



x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
2	$2 - 5 = -3$	$(-3)^2 = 9$
4	$4 - 5 = -1$	$(-1)^2 = 1$
4	-1	1
4	-1	1
5	$5 - 5 = 0$	$0^2 = 0$
5	0	0
7	$7 - 5 = +2$	$(+2)^2 = 4$
9	$9 - 5 = +4$	$(+4)^2 = 16$
Σ	= 0 ✓	= 32

STEP-BY-STEP RESULTS

Step 1 — Mean: $\bar{x} = (2+4+4+4+5+5+7+9)/8 = 40/8 = 5$

Step 2 — Sum of squared deviations: $\Sigma(x_i - \bar{x})^2 = 32$

Step 3 — Variance: $\sigma^2 = 32/8 = 4$

Step 4 — Std Dev: $\sigma = \sqrt{4} = 2$

✓ Interpretation: On average, each value is 2 units away from the mean of 5.



Note: deviations sum to ZERO — squaring prevents cancellation!

RANGE & INTERQUARTILE RANGE (IQR)

Measuring spread using positions in the data

[RANGE]

Range = Maximum – Minimum

The distance between the largest and smallest values.

Simple to compute but very sensitive to outliers — one extreme value inflates it dramatically.

Example: {3, 7, 7, 7, 19}

Range = 19 – 3 = 16

[INTERQUARTILE RANGE (IQR)]

IQR = Q3 – Q1

Q1 = 25th percentile (lower quartile)
Q3 = 75th percentile (upper quartile)

Contains the middle 50% of data.
Resistant to outliers — used in box plots.

Outlier rule: any value below Q1–1.5×IQR or above Q3+1.5×IQR

Metric	Formula	Outlier Sensitive?	Use When
Range	Max – Min	YES — very	Quick, rough estimate
IQR	Q3 – Q1	NO — robust	Data has outliers / skewed
Std Dev	√Variance	Moderate	Normal/symmetric data

QUARTILE CALCULATIONS — STEP BY STEP

How to find Q1, Q2 (Median), Q3, and IQR

WORKED EXAMPLE: Dataset = {4, 7, 9, 12, 15, 18, 22, 25} (n = 8, already sorted)

01

Identify n

n = 8 (even number of observations)
→ Use the average-of-two-middles rule for the median

02

Find Q2 (Median)

Middle two values = position 4 and 5 = 12 and 15
 $Q2 = (12 + 15) / 2 = 13.5$

03

Find Q1 (Lower half)

Lower half = {4, 7, 9, 12} (4 values)
 $Q1 = \text{median of lower half} = (7 + 9) / 2 = 8$

04

Find Q3 (Upper half)

Upper half = {15, 18, 22, 25} (4 values)
 $Q3 = \text{median of upper half} = (18 + 22) / 2 = 20$

05

Calculate IQR

$IQR = Q3 - Q1 = 20 - 8 = 12$
Outlier fences: Lower = $8 - 1.8 = -10$,
Upper = $20 + 1.8 = 38$

Q1

Q2 (Median)

Q3

IQR

Range

8

13.5

20

12

$25 - 8 = 17$

BOX PLOT

A five-number summary — visualising spread and outliers at a glance

A box plot divides data into four sections, each containing approximately 25% of the observations. It provides an instant visual summary of: median, spread, skewness, and outliers.

Minimum

Lowest value within $1.5 \times \text{IQR}$ of Q1 (lower whisker end)

Q1 (25th %ile)

25% of data falls below this value — left edge of box

Median (Q2)

Middle value — line inside the box

Q3 (75th %ile)

75% of data falls below — right edge of box

Maximum

Highest value within $1.5 \times \text{IQR}$ of Q3 (upper whisker end)

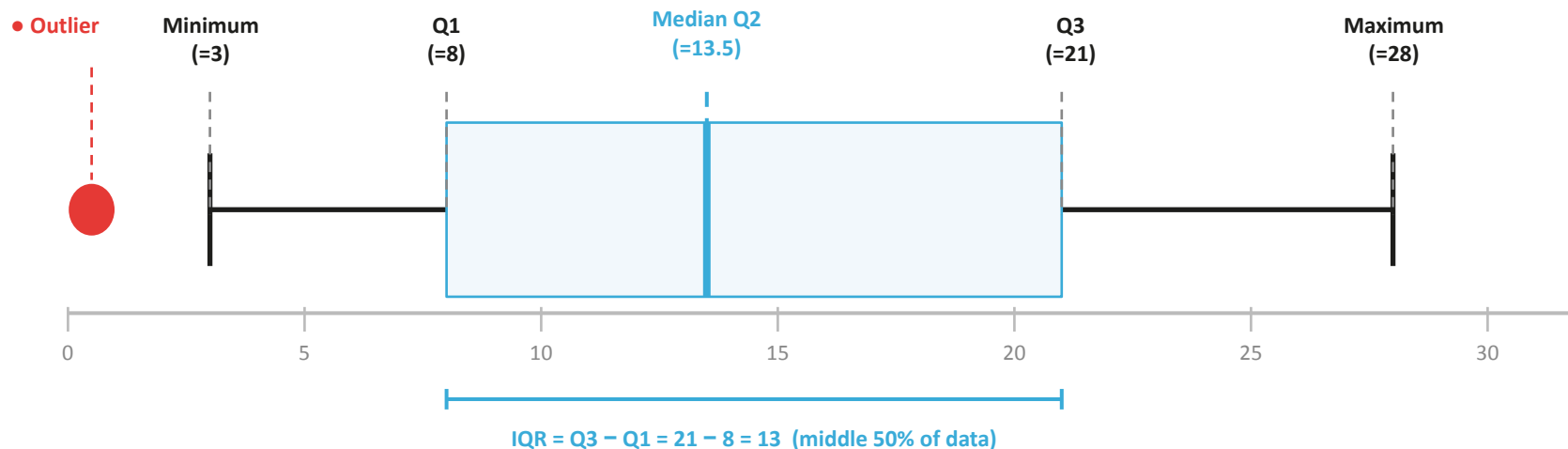
Outliers

Points beyond the whiskers — shown as individual dots

BOX PLOT — FULLY LABELLED DIAGRAM

Reading every element of a box plot at a glance

Dataset: {3, 7, 8, 12, 13, 14, 18, 21, 25, 28} | $Q1=8$ $Q2(\text{Median})=13.5$ $Q3=21$ $IQR=13$ $\text{Min}=3$ $\text{Max}=28$



- The BOX spans Q1 to Q3 — contains the middle 50% of all values.
- WHISKERS extend to the smallest and largest values WITHIN $1.5 \times IQR$ of the box edges.
- OUTLIERS are individual dots plotted BEYOND the whisker ends.

PRACTICE EXERCISES

SECTION 04 — SPREAD & VARIABILITY



Exercise 1 — Calculate Variance & Standard Deviation

Dataset: {6, 8, 10, 12, 14} ($n = 5$)

- a) Compute the mean b) Compute the variance (use population formula) c) Compute the standard deviation d) Interpret: what does the std dev tell you about this dataset?

Exercise 2 — Range and IQR

Dataset (already sorted): {3, 7, 8, 12, 13, 14, 18, 21, 25, 28}

- a) Calculate the Range
b) Find Q1 (25th percentile) and Q3 (75th percentile)
c) Calculate the IQR
d) Determine if 28 is a statistical outlier using the $1.5 \times \text{IQR}$ rule

Exercise 3 — Spread Comparison

Two classes took the same exam:

Class A scores: {70, 71, 69, 72, 68, 70, 71, 70} Mean ≈ 70

Class B scores: {40, 55, 70, 85, 95, 60, 75, 90} Mean ≈ 71

Without calculating: which class has higher variance? What does this mean for a teacher?

SOLUTIONS (1 of 2)

SECTION 04 — SPREAD & VARIABILITY

✓ ANSWERS

✓ Solution 1 — Variance & Standard Deviation

Step 1: Step 1: Mean = $(6+8+10+12+14)/5 = 50/5 = 10$

Step 2: Step 2: Deviations²: $(6-10)^2=16$, $(8-10)^2=4$, $(10-10)^2=0$, $(12-10)^2=4$, $(14-10)^2=16$

Step 3: Step 3: Variance $\sigma^2 = (16+4+0+4+16)/5 = 40/5 = 8$

Step 4: Step 4: Std Dev $\sigma = \sqrt{8} \approx 2.83$

Step 5: Interpretation: On average, each value is about 2.83 units away from the mean of 10 — the data is moderately spread.

SOLUTIONS (2 of 2)

SECTION 04 — SPREAD & VARIABILITY

✓ ANSWERS

✓ Solution 2 — Range and IQR

Step 1: Range = $28 - 3 = 25$

Step 2: $n=10$: Q1 = median of lower half {3,7,8,12,13} = 8. Q3 = median of upper half {14,18,21,25,28} = 21

Step 3: IQR = $Q3 - Q1 = 21 - 8 = 13$

Step 4: Outlier rule: Upper fence = $Q3 + 1.5 \times \text{IQR} = 21 + 19.5 = 40.5$. $28 < 40.5 \rightarrow 28$ is NOT an outlier ✓

✓ Solution 3 — Spread Comparison

Step 1: Class A: Scores cluster tightly around 70 (range ≈ 4) $\rightarrow$ very LOW variance. Students perform consistently.

Step 2: Class B: Scores spread from 40 to 95 (range = 55) $\rightarrow$ HIGH variance. Performance is inconsistent.

Step 3: For a teacher: Class A is uniformly skilled; Class B has both struggling students (40,55) and high-achievers (90,95) — may need differentiated instruction.

SECTION 05

DISTRIBUTIONS

Normal, Skewed, Bimodal, Logarithmic — the shapes data can take

NORMAL DISTRIBUTION

The bell curve — the most important distribution in statistics

[NORMAL / GAUSSIAN DISTRIBUTION]

A symmetric, continuous distribution where most values cluster near the mean and frequencies decrease as you move to either tail. Fully described by two parameters: mean (μ) and std dev (σ).

Symmetry

Mean = Median = Mode. The distribution is perfectly mirrored around its centre.

68-95-99.7 Rule

68% of data within $\pm 1\sigma$. 95% within $\pm 2\sigma$. 99.7% within $\pm 3\sigma$.

Continuous

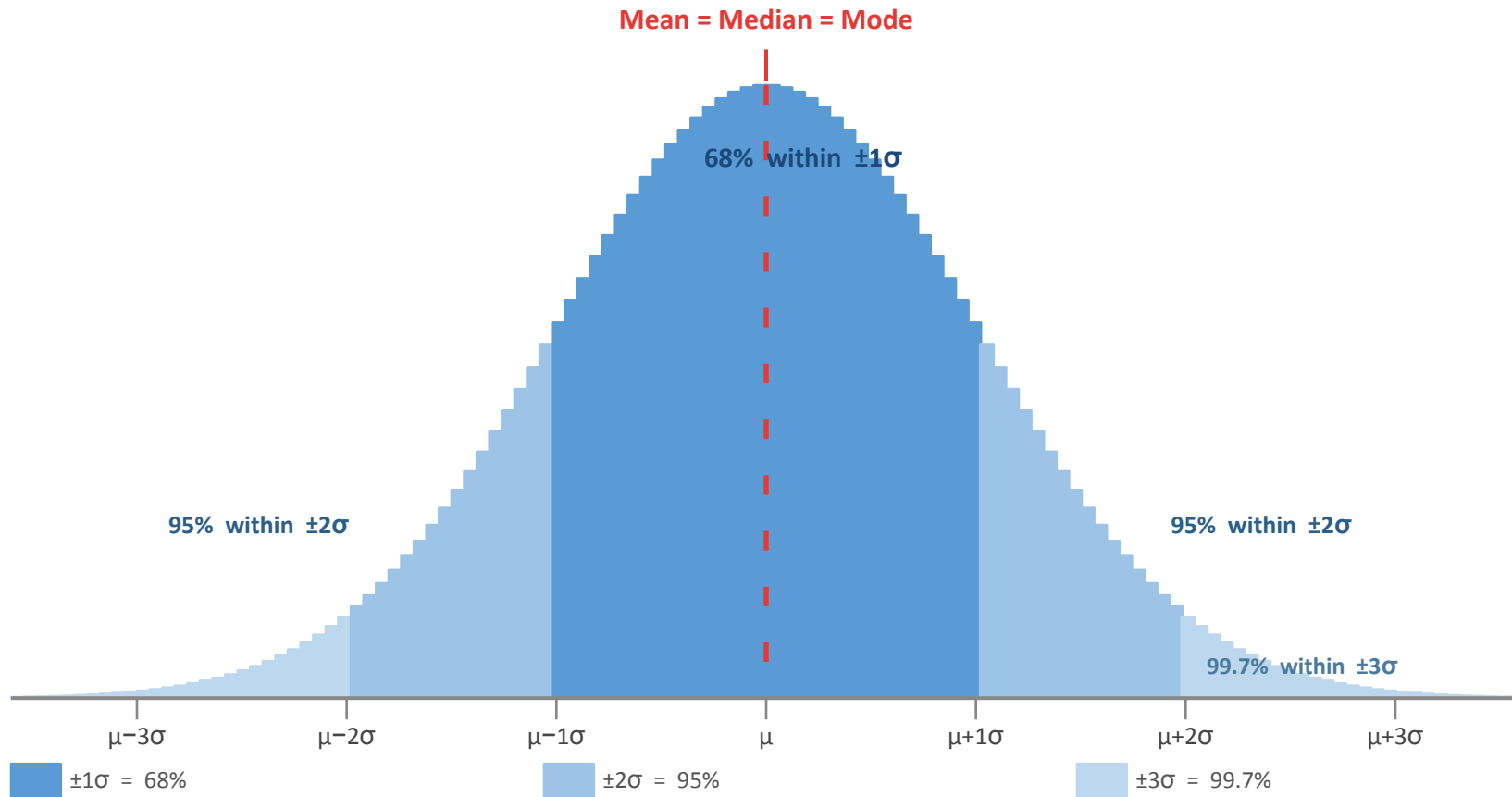
No gaps — values can take any real number. The tails approach but never reach zero.

Outlier Detection

Values beyond $\pm 2\sigma$ or $\pm 3\sigma$ are considered unusual — used for anomaly detection.

NORMAL DISTRIBUTION — BELL CURVE

The 68–95–99.7 rule and why Mean = Median = Mode



SKEWED DISTRIBUTIONS

When the tail pulls the mean away from the median

Unlike a bell curve, in a skewed distribution the mean, median, and mode are NOT the same. The long tail skews (pulls) the mean in its direction.

POSITIVE SKEW (Right-Skewed)

Long tail extends to the RIGHT.

Mode < Median < Mean
(Mean is pulled highest by extreme right values)

Examples:

- Income distribution (most earn little, few earn millions)
- House prices
- Time to complete a task

The mean overstates typical value → use Median.

NEGATIVE SKEW (Left-Skewed)

Long tail extends to the LEFT.

Mean < Median < Mode
(Mean is pulled lowest by extreme left values)

Examples:

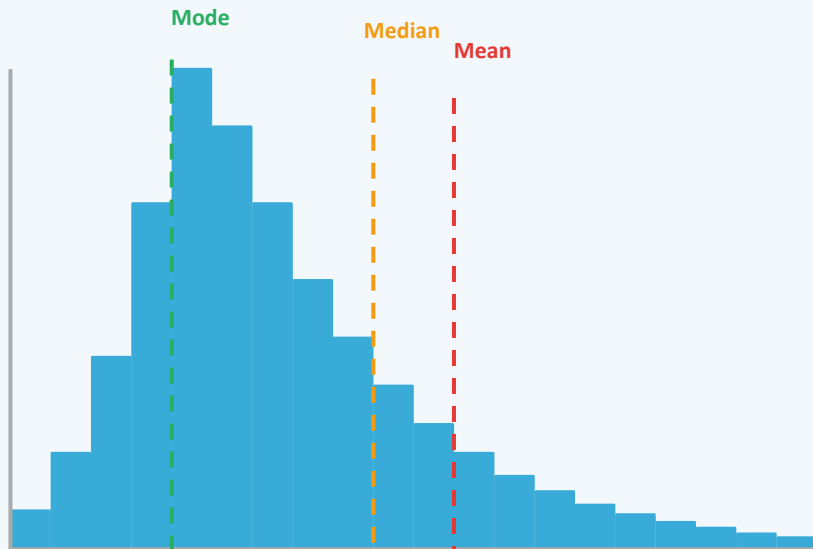
- Retirement age (most retire late, few retire early)
- Test scores when exam is easy
- Age at death in developed countries

The mean understates typical value → use Median.

SKewed DISTRIBUTIONS

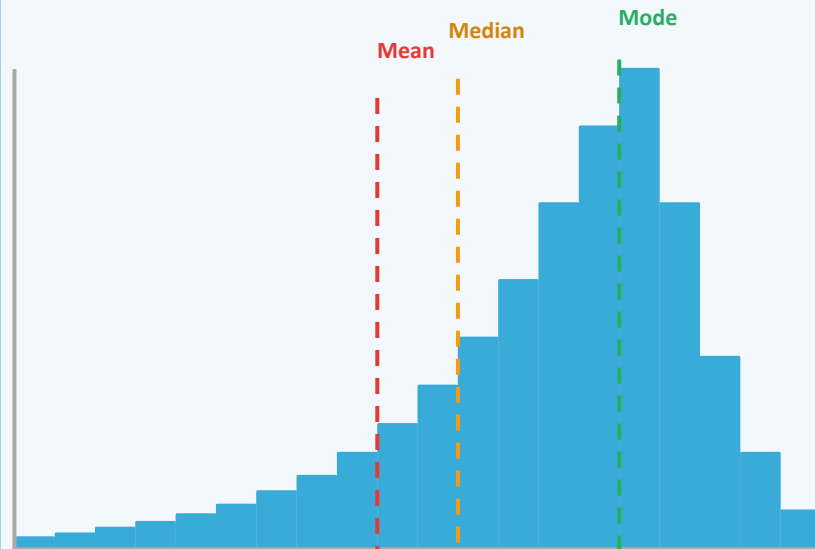
Direction of the tail determines where Mean, Median, Mode fall

► POSITIVE SKEW (Right-Skewed)



Mode < Median < Mean
Long tail → RIGHT
Mean pulled toward tail
Examples: income, house prices

◀ NEGATIVE SKEW (Left-Skewed)



Mean < Median < Mode
Long tail → LEFT
Mean pulled toward tail
Examples: retirement age, easy exams

OTHER DISTRIBUTIONS

Bimodal, Multimodal, Logarithmic, and Triangular

BIMODAL / MULTIMODAL

TWO (or more) peaks occur.

Suggests the data may be a mix of two sub-groups.

Solution: split data into groups and analyse each.

Example: heights of a mixed male+female sample.

LOGARITHMIC (Log-Normal)

Almost all values clustered on one side with a long tail.

Common in wealth, income, and web traffic data.

Take log of data to convert to a normal distribution.

Example: distribution of wealth / social media followers.

TRIANGULAR

Simple shape: straight lines between Min, Mode, Max.

Used for PERT project time estimation when only 3 values known (optimistic, most likely, pessimistic).

Example: task completion time estimates.

UNIFORM

Every value has an equal probability — completely flat.

No peaks, no skew.

Example: rolling a fair die — each number (1–6) equally likely.

Used in random number generation.

BIMODAL, LOGARITHMIC & UNIFORM DISTRIBUTIONS

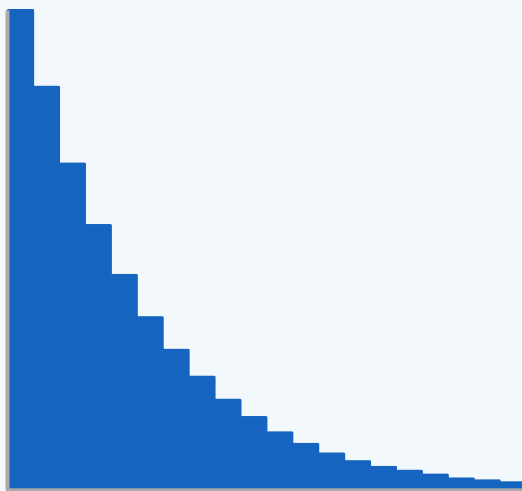
Recognising other common data shapes

BIMODAL



Two distinct peaks → two sub-groups.
Split and analyse separately.
Eg: heights of mixed male+female sample.

LOGARITHMIC



Values clustered low, long right tail.
Apply log transform → normal.
Eg: wealth, social media follower counts.

UNIFORM



All values equally likely. Flat shape.
No peak or skew.
Eg: fair die roll, random number gen.

PRACTICE EXERCISES

SECTION 05 — DISTRIBUTIONS



Exercise 1 — Identify the Distribution

For each description, name the most likely distribution shape:

- a) SAT scores of thousands of students cluster near 1000, tapering symmetrically on both sides.
- b) Annual salaries in a company: most earn \$30k–60k, a CEO earns \$5M.
- c) A survey of gym-goers shows two peaks — one for people who go 1x/week and one for 5x/week.
- d) Rolling a fair 6-sided die 10,000 times.

Exercise 2 — Skewness & Central Tendency

For a positively skewed dataset representing household incomes:

- a) Which measure is largest: Mean, Median, or Mode?
- b) Which measure best represents the 'typical' household income?
- c) A journalist reports the 'average' income using the mean — is this misleading? Why?

Exercise 3 — 68-95-99.7 Rule

IQ scores follow a normal distribution with $\mu = 100$ and $\sigma = 15$.

- a) What % of people have IQ between 85 and 115?
- b) What % of people have IQ between 70 and 130?
- c) What IQ range covers 99.7% of the population?

✓ Solution 1 — Identify the Distribution

Step 1: a) SAT scores symmetric around 1000 → NORMAL distribution (bell curve)

Step 2: b) Most earn \$30k–60k but CEO earns \$5M (extreme right outlier) → POSITIVE SKEW (right-skewed / logarithmic)

Step 3: c) Two peaks (1x/week and 5x/week) → BIMODAL distribution — likely two sub-groups (casual vs dedicated gym-goers)

Step 4: d) Fair die: all 6 outcomes equally likely → UNIFORM distribution

✓ Solution 2 — Skewness & Central Tendency

Step 1: a) In positive skew: Mode < Median < Mean → The MEAN is the largest value

Step 2: b) MEDIAN best represents typical household income — it is not affected by the millionaires at the top

Step 3: c) YES, it's misleading. The mean is inflated by high earners (e.g. a few billionaires). Reporting mean makes typical income look much higher than it actually is for most people.

✓ Solution 3 — 68-95-99.7 Rule

Step 1: a) $85 \text{ to } 115 = \mu \pm 1\sigma$ (100 ± 15) → 68% of people have IQ in this range

Step 2: b) $70 \text{ to } 130 = \mu \pm 2\sigma$ (100 ± 30) → 95% of people have IQ in this range

Step 3: c) $\mu \pm 3\sigma = 100 \pm 45$ → range: 55 to 145 covers 99.7% of the population

SECTION 06

SUMMARY & APPLICATIONS

Everything together — the complete descriptive statistics toolkit

COMPLETE CHEAT SHEET

All formulas and definitions at a glance

Concept	Formula / Rule	When to Use
Mean $\bar{X}$	$\sum x_i / n$	Symmetric data, no outliers
Median	Middle value (sorted)	Skewed data or outliers present
Mode	Most frequent value	Categorical data or multimodal
Range	Max – Min	Quick spread estimate
Variance σ^2	$\sum (x_i - \bar{X})^2 / N$ (or $n-1$ for sample)	Measure dispersion (squared units)
Std Deviation σ	$\sqrt{\text{Variance}}$	Dispersion in original units
IQR	$Q3 - Q1$	Robust spread; used in box plots
Outlier Fence	$Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$	Identify outliers
Rel. Frequency	$\text{freq} / n \times 100\%$	Proportion of each value
Normal 68-95-99.7	$\pm 1\sigma = 68\%$ $\pm 2\sigma = 95\%$ $\pm 3\sigma = 99.7\%$	Normal distribution reference

DESCRIPTIVE ANALYSIS WORKFLOW

A step-by-step guide to analysing any new dataset

01

Understand Variable Types

Identify each variable:
Numeric or Categorical?
Continuous or Discrete?

02

Build Frequency Table

For discrete/categorical:
list values, count
frequencies,
compute relative
frequencies.

03

Compute Central Tendency

Calculate Mean, Median,
Mode.
Choose the right one
based
on data shape and outliers.

04

Measure Spread

Compute Range, IQR,
Variance,
and Std Deviation.
Draw a box plot to
visualise.

05

Identify Distribution

Check symmetry /
skewness.
Label the distribution
shape.
Note any outliers.

APPLIED EXAMPLE 01 | Standard Deviation in Business

Two Offices, Same Revenue — Different Story

$$\sigma_1 = \$2K$$

$$\sigma_2 = \$25K$$

Why the mean alone can be dangerously misleading

*A SaaS company has two regional sales offices.
Both report the same monthly recurring revenue.*

The Numbers — What Are You Looking At?

Scenario | Both offices report \$50,000 avg monthly revenue



Same average revenue. Same target. Same team size. → Should your management decisions be the same?



Office A — Stable

Mean (μ)	\$50,000 / month
Std Dev (σ)	\$2,000
Typical range ($\pm 1\sigma$)	\$48K → \$52K

Monthly revenue fluctuation:



 **Tight band** — forecasting is accurate and reliable

Low variance = low noise = easy to model



Office B — Volatile

Mean (μ)	\$50,000 / month
Std Dev (σ)	\$25,000
Typical range ($\pm 1\sigma$)	\$25K → \$75K

Monthly revenue fluctuation:



 **Wide band** — the \$50K average is misleading here

High variance = high noise = hard to model



What Happened → What It Means → What You Do

Turning σ into management decisions

 The Situation	 Statistical Meaning	 Management Action	 The Benefit
 You read the data: Office A: $\sigma = \$2K$ Office B: $\sigma = \$25K$	Office A is stable — revenue always near \$50K. Office B is chaotic — one month \$25K, next \$75K!	Investigate Office B across time: Is the spike tied to quarters? Specific clients? Seasonality?	You find the root cause instead of just trusting the average
 You're setting headcount and team shifts for both offices	Office A: demand is predictable — schedule with confidence. Office B: you can't predict — feast or famine months.	Office A: fixed team size Office B: flexible contracts or on-call coverage	Cut operational costs and avoid burnout or idle time
 You're deciding the budget allocated to each office	Office A: reliable — every dollar allocated will be used efficiently. Office B: risky — budget may go unused or be exceeded.	Office A: fixed monthly budget Office B: base budget + contingency reserve ($\pm 30\%$)	Reduce financial risk and improve quarterly P&L accuracy

Why This Matters for Data Scientists

From the SaaS office example to real ML problems

Same mean, different σ = two completely different modelling challenges



Variance = Data Noise

High variance means the signal is buried in noise.

Office B is hard to model because there's no stable pattern to learn — every month looks different to your algorithm.



σ = Your Uncertainty Radius

In a normal distribution, 68% of data falls within $\pm 1\sigma$.

Office A: 68% of months fall between \$48K–\$52K

Office B: 68% of months fall between \$25K–\$75K

→ completely different planning horizon.



Impact on ML Models

Feature scaling (Z-score): $z = (x - \mu) / \sigma$

If you skip scaling on Office B data, the wide value range will dominate your model and distort predictions for all other features.



Mean Alone Is Misleading

\$50K average for Office B isn't wrong — it's incomplete.

Like saying: 'The average depth of this river is 1m' — but some spots are 10cm and others are 3m.

You'd still drown planning around just the average.

APPLIED EXAMPLE 02 | Central Tendency + Spread

10 Sales Reps, One Quarter of Deals

Every measure of central tendency and spread — grounded in real sales data

Mean · Median · Mode · σ · IQR

Sales Reps Q3 Performance — Measures of Center

Where is the 'typical' performance? Three different answers.

 Monthly deals closed by 10 sales reps (\$K): 60 · 65 · 70 · 70 · 75 · 80 · 85 · 90 · 95 · 100

1
Mean
(Average)

Formula: $(60+65+70+70+75+80+85+90+95+100) \div 10 = 790 \div 10 = \$79K$

 **What happened:** The team average is \$79K per rep this quarter.

What it means: If we redistributed all deals equally, each rep gets \$79K.
Sensitive to outliers — that \$100K rep pulls it up.

\$79K

2
Median
(Middle)

Sorted: 60, 65, 70, 70, 75, 80, 85, 90, 95, 100 → two middle values: 75 & 80 → $(75+80) \div 2 = \$77.5K$

 **What happened:** Half the reps closed below \$77.5K, half above.

What it means: More reliable than mean when top performers skew the data.
Use this for realistic benchmarking.

\$77.5K

3
Mode
(Most common)

Mode = \$70K — the only value that appears twice (two reps hit exactly this number)

 **What it means:** \$70K is your most typical individual performance — use it as the baseline target when onboarding new reps or setting minimum expectations.

\$70K

Sales Reps Q3 — Measures of Spread

How scattered is the performance across the team?

Data: 60 · 65 · 70 · 70 · 75 · 80 · 85 · 90 · 95 · 100 (\$K) | Mean = \$79K

1 Range

$$\text{\$100K} - \text{\$60K} = \text{\$40K}$$

📌 A \$40K gap between your best and weakest rep

⚙️ Flag reps below \$65K for a performance review

⚡ Sensitive to outliers — one exceptional rep changes this entirely

2 Variance

$$\Sigma(x - 79)^2 \div 10 \approx \text{\$170 (K}^2\text{)}$$

📌 Average squared deviation from the mean = 170

⚙️ High number → team performance is scattered → review coaching consistency

⚡ Squared units — can't compare directly with the revenue numbers

3 Standard Deviation (σ)

$$\sqrt{170} \approx \text{\$13K}$$

📌 Most reps perform within $\pm$ \$13K of the \$79K average

⚙️ Expected normal range: \$66K–\$92K
Any rep outside this band → investigate (star or at-risk)

✅ Same unit as the data — most interpretable spread measure

4 Interquartile Range (IQR)

$$Q1=\text{\$70K}, Q3=\text{\$90K} \rightarrow \text{IQR} = \text{\$20K}$$

📌 The middle 50% of reps close between \$70K–\$90K

⚙️ Rep below \$55K or above \$105K → statistical outlier → flag for review

✅ Resistant to outliers — best for skewed or uneven distributions

The Complete Picture — Every Number, Every Decision

Connecting each statistical measure to a real management action

 Measure & Value	 What It Tells You	 Action You Take	 Why It Matters
 Mean = \$79K Average	Overall team performance level this quarter	Use as the primary KPI on dashboards and exec reports	Fast objective benchmark for any rep
 Median = \$77.5K Middle value	True 50th percentile — less pulled by top earner	Use for planning when 1–2 reps are exceptional outliers	More realistic baseline for average-level expectations
 Mode = \$70K Most common	The most frequently achieved performance level	Set as the minimum onboarding target for new reps	Grounds expectations in what most people actually achieve
 σ = \$13K Std Deviation	Most reps fall within \$66K–\$92K ($\pm 1\sigma$ band)	Any rep outside this band: investigate (star or at-risk)	Early warning system — no need to wait for end-of-quarter
 IQR = \$20K Middle 50%	Core of the team is between \$70K and \$90K	Below \$55K or above \$105K → outlier → escalate review	Robust detection unaffected by extreme performers

What This Means When You Build Models

Every measure you calculated has a direct ML application

The sales rep data isn't just a classroom exercise — it's the exact kind of HR analytics dataset you'd work with as a Data Scientist.



Feature Scaling

$\sigma = \$13\text{K}$ feeds directly into Z-score normalisation:

$$z = (x - \mu) / \sigma \rightarrow z = (x - 79) / 13$$

Without this step, the \$100K rep's data dominates every model trained on this dataset.



Outlier Detection

Reps outside $\pm 2\sigma$ (below \$53K or above \$105K) are statistical outliers.

In an anomaly detection pipeline, σ and IQR are the two standard tools to flag these automatically — no manual review needed.



Distribution Check

Mean (\$79K) $\approx$ Median (\$77.5K)
→ Near-symmetrical distribution → safe to apply Gaussian models.

If mean $\gg$ median, you'd need to log-transform the data first.



Prediction Confidence

High variance in a feature → the model struggles to find a stable pattern.

If Office B's revenue data entered your forecast model, prediction intervals would be huge — low business value output.



Hypothesis Testing

Is a new rep's \$58K performance genuinely below average, or just noise?

$$\text{Z-test: } z = (58 - 79) / 13 = -1.6$$

At -1.6σ , it's borderline — worth a coaching conversation, not a dismissal.



Business Dashboards

Each metric maps to a dashboard element:

- Mean → KPI headline number
- Median → realistic team benchmark
- IQR → box plot for performance spread
- σ → confidence band on forecast charts

PRACTICE EXERCISES

SECTION 06 — SUMMARY & APPLICATIONS

 TRY IT

Exercise 1 — Full Analysis — End-to-End

Dataset: Monthly salaries (EGP thousands) of 10 employees:

{4, 5, 5, 6, 7, 7, 7, 8, 9, 40}

a) Compute Mean, Median, Mode b) Compute Range and IQR c) Is the value '40' an outlier? (Use $1.5 \times \text{IQR}$ rule) d) Which measure best represents a typical salary? Why?

Exercise 2 — Distribution Identification & Implication

After analysis, you find: Mean = 11.4k, Median = 7k, Mode = 7k

- a) What type of distribution does this suggest?
- b) Sketch (describe) what a histogram of this data would look like.
- c) What would you report to HR as the 'typical' salary?

Exercise 3 — Variable Classification + Table

A hospital records: Patient age, Blood type, Pain scale (1–10), Days admitted (whole numbers), Recovery status (full/partial/none).

For each variable: a) State the type b) Name the best measure of central tendency c) Name the best measure of spread.

SOLUTIONS (1 of 2)

SECTION 06 — SUMMARY & APPLICATIONS

✓ Solution 1 — Full End-to-End Analysis

Step 1: Sorted: {4,5,5,6,7,7,7,8,9,40} $n=10$

Step 2: Mean = $(4+5+5+6+7+7+7+8+9+40)/10 = 108/10 = 10.8k$ | Median = $(7+7)/2 = 7k$ | Mode = 7k (appears 3 times)

Step 3: Range = $40-4 = 36k$ | $Q1=5$, $Q3=8$, $IQR=3k$

Step 4: Outlier check: Upper fence = $Q3+1.5 \times IQR = 8+4.5 = 12.5k$. $40 > 12.5 \rightarrow$ YES, 40k is a statistical OUTLIER ✓

Step 5: Best measure: MEDIAN (7k). The outlier (40k) inflates the mean to 10.8k — far above what 9/10 employees actually earn.

SOLUTIONS (2 of 2)

SECTION 06 — SUMMARY & APPLICATIONS

✓ ANSWERS

✓ Solution 2 — Distribution & Implication

Step 1: Mean (10.8) > Median (7) = Mode (7) → This is a POSITIVE SKEW (right-skewed) distribution

Step 2: Histogram: A peak around 7k on the left, with a long thin tail stretching far to the right toward 40k

Step 3: Report to HR: The MEDIAN salary of 7k EGP — this accurately represents what a typical employee earns without being distorted by the executive's 40k salary

✓ Solution 3 — Variable Classification

Step 1: Patient age → Numeric-Continuous | Best central tendency: Mean (if symmetric) or Median (if skewed) | Best spread: Std Dev or IQR

Step 2: Blood type (A/B/AB/O) → Categorical-Nominal | Best central tendency: MODE | Spread: N/A (use frequency table)

Step 3: Pain scale (1–10) → Categorical-Ordinal | Best central tendency: MEDIAN (ranked scale) | Spread: IQR

Step 4: Days admitted (whole #) → Numeric-Discrete | Best central tendency: Mean or Median | Spread: Std Dev or IQR

Step 5: Recovery status (full/partial/none) → Categorical-Ordinal | Best central tendency: MEDIAN or MODE | Spread: IQR

WHAT'S NEXT?

YOU NOW HAVE THE TOOLS

Variables • Frequency Tables • Mean / Median / Mode • Variance • Std Dev • IQR • Box Plots • Distributions

COMING UP

Inferential Statistics

Confidence intervals,
hypothesis testing

Probability Theory

Conditional probability,
Bayes' theorem

ML Foundations

How statistics powers
machine learning models

Thank you for your attention

[i]NSTANT